

Enhancing Anti-Interference of Magnetic Tracking: A MagRobustNet-Based Framework With Self-Supervised Anomaly Detection and Measurements Recovery

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Abstract—Magnetic tracking technology shows great promise for applications in medicine and industry. However, it often suffers from diverse and unpredictable interferences in practical applications, such as hard-/soft-iron interferences and sensor saturation, leading to reduced localization accuracy or even tracking failure. Thus, we propose a two-step framework to mitigate the impact of interferences based on MagRobustNet, a UNet-like autoencoder network. In the first step, disjoint mask sets are used in conjunction with MagRobustNet to detect anomalous measurements subject to disturbances. In the second step, the interfered regions are masked, and MagRobustNet is applied again to recover their expected measurements from neighboring normal data. Experimental results from testing in four interference scenarios showed that the proposed method improved the average position accuracy by 76.2%, enhancing the tracking system's anti-interference capability. In addition, the proposed method can indicate the interfered regions, thereby prompting the adjustment of the magnetometer array to an interference-free location and offering a new potential diagnostic method for localizing ingested foreign bodies in clinical practice.

Index Terms—Anomaly detection, anti-interference, magnetic sensing, measurement recovery, robust tracking.

I. INTRODUCTION

MAGNETIC tracking is a low-cost, occlusion-free, and high-precise technique with great potential for numerous applications, including wireless capsule endoscopy (WCE) localization [1], flexible robot shape sensing [2], automated guided vehicle (AGV) navigation [3], and body motion capture [4]. However, magnetic tracking systems (MTSs) are sensitive to ferromagnetic materials, which will result in degraded localization accuracy or even tracking failure. Thus, it is crucial to enhance the tracking system's anti-interference ability and robustness to avoid medical malpractices and financial losses.

The factors affecting tracking robustness can be classified as related to the tracking algorithm or environmental interferences. Traditional MTSs are mainly based on the Levenberg–Marquardt (LM) optimization algorithm [5], [6], [7], and therefore are sensitive to initial guesses. To address this issue, some researchers attempted to provide initial guesses for the LM algorithm using various initial estimation methods, such as particle swarm optimization (PSO) [8], whale optimization algorithm (WOA) [9], artificial neural networks (ANN) [10], while others used end-to-end neural networks to directly regress the magnet pose [11], [12]. Among these methods, AMagPoseNet [12] demonstrated superior localization accuracy and computation latency compared to traditional methods and is also independent of initial guesses.

Environment interferences in MTSs arise primarily from hard and soft iron effects [13]. The hard iron effects are caused by objects that can generate magnetic fields, such as permanent magnets and electrified wires. The soft iron effect results from materials that distort the magnet's magnetic field, such as ferromagnetic surgical instruments or iron plates. Fig. 1(a) shows the potential interference sources presented in the operating room. In addition, previous research has identified a reduction in localization accuracy at lower heights due to sensor saturation, as depicted in Fig. 1(b).

Sensor-level anti-interference: To address the impact of soft-/hard-iron interferences caused by currents and ferromagnetic materials in the environment, Wu et al. [13] employed the

Received 7 August 2024; accepted 10 September 2024. Date of publication 17 October 2024; date of current version 7 January 2025. This work was supported in part by the Hong Kong Research Grants Council through the Collaborative Research Fund under Grant CRF C4063-18G, in part by the General Research Fund under Grant GRF 14216022, in part by the NSFC/RGC Joint Research Scheme 2022/23 N_CUHK420/22, in part by the National Academy of Medicine, Healthy Longevity Catalyst Awards (Hong Kong), in part by the Chinese University of Hong Kong Mainland Postdoctoral Sponsorship Program, and in part by the Shenzhen-Hong Kong-Macau Technology Research Program (Type C) under Grant 202108233000303. Paper no. TII-24-4005. (Corresponding author: Hongliang Ren.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Animal Ethics Committee of Qilu Hospital of Shandong University under Application No. Dwll-2021-021, and performed in line with the Guiding Principles in the Care and Use of Animals.

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TII.2024.3461774>, provided by the authors.

Digital Object Identifier 10.1109/TII.2024.3461774

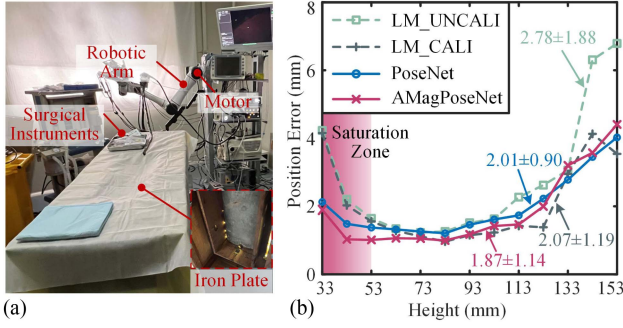


Fig. 1. Factors impacting the robustness of MTS. (a) Potential interference sources presented in the operating room (surgical instruments, robotic arm, iron plate, et al.). (b) Reduction in localization accuracy at low heights due to sensor saturation, reported from [12], where the magnetometer range was set to $\pm 1200 \mu\text{T}$.

optimal maximum likelihood method to estimate the parameters of an error model, while Christensen et al. [14] utilized a neural network to learn a magnetic field distortion model. However, existing techniques primarily focus on magnetic compass applications and rely on precollected calibration data in specific usage environments, rendering them unsuitable for handling diverse and unpredictable disturbances. As for the saturation induced by strong magnetic fields, Fu et al. [15] applied an external electromagnetic coil for interference compensation. Currently, no available research can address the saturation issue solely by post-processing without external hardware.

System-level anti-interference: To enhance the resistance of MTSs to environmental interferences, Su [16] et al. utilized the Huber loss function to mitigate the impact of hard iron interference on localization. Still, this method is limited to traditional optimization-based approaches and cannot identify which magnetometers are affected. Son [6] and Shi [17] et al. introduced an actuator's magnetic field to manipulate WCE with an embedded magnet. They accurately calculated the actuator's magnetic field based on magnetic models and subtracted it from the raw measurements. However, they cannot address unknown stochastic interferences caused by soft-/hard-iron effects and sensor saturation. Song [18] and Shao [19] et al. optimized the sensor layout using the grid method and SNR evaluation, which improved the system's localization accuracy to some extent. Nevertheless, they were also unable to cope with these random and diverse interferences.

System-level anomaly detection: Diverse and unpredictable interferences can cause measured data to deviate from the normal distribution [20]. Anomaly detection is recognized as a means to enhance the robustness of sensing systems by detecting and indicating the presence of interferences. Various anomaly detection techniques are available in sensing systems, including traditional methods, such as clustering [21] and support vector machine [22], deep learning-based approaches, such as autoencoders [23] and generative adversarial networks (GANs) [24], and hybrid methods combining traditional and deep learning, such as deep one-class classification [25]. However, detecting interferences is often insufficient for MTSs, as our goal is to recover their expected measurements and obtain more reliable localization results.

Current research on anti-interference in MTSs is limited because it is challenging to separate the magnetic field generated by the tracking target (permanent magnet) from diverse

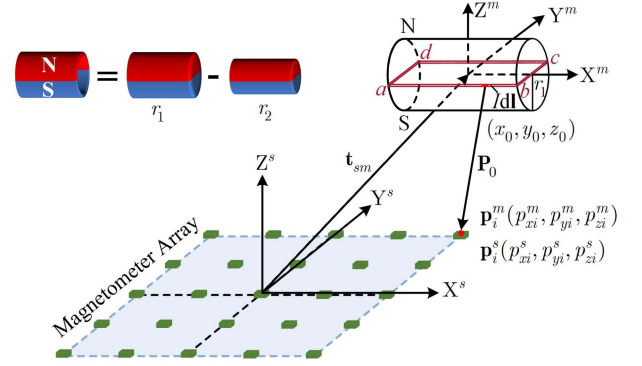


Fig. 2. Modeling an annular magnet based on the equivalent magnetizing current, the Biot-Savart law, and the superposition principle.

unpredictable interference fields. This study is the first to introduce autoencoder networks to tackle this issue. Specifically, we propose a two-step framework with anomaly detection and measurement recovery. The contributions of our framework are summarized as follows.

- 1) Designed a UNet-like autoencoder network, MagRobustNet, with partial convolution to accommodate ultralow resolution (5×5) measurements of MTSs.
- 2) Integrated anomaly detection and measurement recovery into our framework using only MagRobustNet. This not only indicates anomalous regions subject to disturbances but also improves the average position accuracy of MTS in four interference scenarios by 76.2%.
- 3) Presented a model-to-real domain adaptation approach to ensure that our framework can be used for both learning-based and optimization-based tracking methods.

II. MAGNETIC FIELD MODEL

A. Prior Mathematical Model of Annular Magnets

As illustrated in Fig. 2(a), a sensor array composed of N tri-axes magnetometers is adopted to perceive the magnetic induction intensity generated by an annular magnet in space. The i th magnetometer's position in the magnet coordinate frame $\{m\}$ is $\mathbf{p}_i^m = [p_{xi}^m, p_{yi}^m, p_{zi}^m]^T$. Based on the equivalent magnetizing current model and the Biot-Savart law, the magnetic induction intensity $d\mathbf{B}_l$ generated by a surface current element located at (x_0, y_0, z_0) on the i th magnetometer can be expressed as

$$d\mathbf{B}_l = \frac{\mu_0 \mu_r}{4\pi} \frac{I d\mathbf{l} \times \mathbf{P}_0}{|\mathbf{P}_0|^3} = K \frac{d\mathbf{v} \times \mathbf{P}_0}{|\mathbf{P}_0|^3} \quad (1)$$

where μ_0 and μ_r denote the magnetic permeability of vacuum and the relative magnetic permeability of the medium, respectively, and $\mathbf{P}_0$ is the vector from the point (x_0, y_0, z_0) to point $\mathbf{p}_i^m$. The surface current $\mathbf{I} = I d\mathbf{l} = J_s d\mathbf{v}$, where I and $d\mathbf{l}$ represent the current magnitude and direction, and J_s is the current density (A/m). Let $K = \frac{J_s \mu_0 \mu_r}{4\pi}$, which is a constant determined by magnet material.

Assuming that the radius of a cylinder magnet is r_1 . The magnetic induction intensity at $\mathbf{p}_i^m$ can be estimated by

$$\mathbf{B}_{ci}^m(\mathbf{p}_i^m, r_1) = \int_{-r_1}^{r_1} d\mathbf{B}_z \quad (2)$$

where $d\mathbf{B}_z = \int_{ab} d\mathbf{B}_l + \int_{bc} d\mathbf{B}_l + \int_{cd} d\mathbf{B}_l + \int_{da} d\mathbf{B}_l$ is the magnetic induction intensity generated by the surface current $\mathbf{I}_{ab\cdots da}$ at the point $\mathbf{p}_i^m$.

Applying the superposition principle of magnetic fields, the difference between the magnetic induction intensity produced by two cylindrical magnets with different radii yields an equivalent annular magnet with the same magnetic moment direction and length. Then, the magnetic induction intensity at the point $\mathbf{p}_i^m$ due to the annular magnet is given by

$$\mathbf{B}_{ai}^m(\mathbf{p}_i^m, r_1, r_2) = \mathbf{B}_{ci}^m(\mathbf{p}_i^m, r_1) - \mathbf{B}_{ci}^m(\mathbf{p}_i^m, r_2) \quad (3)$$

where r_1 and r_2 denote its outer and inner radii, respectively.

Given the magnet pose $\xi = [x, y, z, \alpha, \beta, \gamma]$ in the system coordinate frame $\{s\}$, which describes the translation $\mathbf{t}_{sm} = [x, y, z]^T$ and rotation $\mathbf{R}_{sm} = \mathbf{R}_x(\alpha)\mathbf{R}_y(\beta)\mathbf{R}_z(\gamma)$ of the frame $\{m\}$ relative to $\{s\}$. The i th magnetometer's position in the frame $\{s\}$ is $\mathbf{p}_i^s = [p_{xi}, p_{yi}, p_{zi}]^T$, and the magnetic induction intensity at $\mathbf{p}_i^s$ can be calculated by

$$\begin{aligned} \mathbf{B}_{ai}^s &= \mathbf{R}_{sm} \mathbf{B}_{ai}^m(\mathbf{p}_i^m, r_1, r_2) \\ &= \mathbf{R}_{sm} \mathbf{B}_{ai}^m(\mathbf{R}_{gm}^{-1} \cdot (\mathbf{p}_i^s - \mathbf{t}_{gm}), r_1, r_2) \\ &= \mathbf{F}(\mathbf{p}_i^s, \xi), (i = 1, 2, \dots, N). \end{aligned} \quad (4)$$

The prior model (4) enables predicting the measurements of the magnetometer array for any given magnet pose.

B. Model-to-Real Domain Adaptation

Training MagRobustNet requires a substantial amount of normal data. We can employ the prior model (4) to generate a comprehensive and sufficient dataset, avoiding the laborious process of manual data collection. However, a domain shift arises between the generated and real-world data, primarily attributed to several factors, including magnetometer measurement errors, manufacturing and welding errors in the magnetometer array, and manufacturing errors in the magnets.

To mitigate the impact of these errors on localization, traditional optimization-based methods introduce a calibration procedure [7], while deep learning-based methods, such as AMagPoseNet, design a CaliNet to facilitate the transfer from the real-world domain to the mathematical model domain, known as real-to-model domain adaptation [12].

In this study, we present a method for data generation by a precalibration operation, which enables model-to-real domain adaptation. Assuming the real measured sensor data of the i th magnetometer is denoted as $\tilde{\mathbf{B}}_{ai}^s$, a total of L real measurement samples are used for calibration. The magnetic induction intensity, $\mathbf{B}_{ai}^s$, is a function related to the constant K , i.e., $\mathbf{B}_{ai}^s = K\hat{\mathbf{B}}_{ai}^s$. We define the objective function related to K as

$$E_K = \sum_{j=1}^L \left\| \tilde{\mathbf{B}}_{ai}^s(j) - K\hat{\mathbf{B}}_{ai}^s(j) \right\|_2^2. \quad (5)$$

Minimizing E_K through the linear least squares method yields a calibrated parameter $\tilde{K}$ specific to the actual magnet, i.e.,

$$\tilde{K} = \frac{\sum_{i=1}^L \tilde{\mathbf{B}}_{ai}^s \hat{\mathbf{B}}_{ai}^s}{\sum_{i=1}^L \hat{\mathbf{B}}_{ai}^s \hat{\mathbf{B}}_{ai}^s}. \quad (6)$$

Similarly, we define the objective function associated with the magnetometer position and orientation as

$$\begin{aligned} E_{pi} &= \sum_{j=1}^L \left\| \tilde{\mathbf{B}}_{ai}^s(j) - \mathbf{B}_{ai}^s(j) \right\|_2^2 \\ E_{oi} &= \sum_{j=1}^L \left\| \tilde{\mathbf{B}}_{ai}^s(j) - \mathbf{O}_i \cdot \mathbf{B}_{ai}^s(j) \right\|_2^2 \end{aligned} \quad (7)$$

where $\mathbf{O}_i \in \mathbb{R}^{3 \times 3}$ represents the orientation compensation matrix of magnetometer. By minimizing E_{pi} and E_{oi} , we can obtain a calibrated magnetometer position $\tilde{\mathbf{p}}_i^s$ and an orientation compensation matrix $\tilde{\mathbf{O}}_i$. Substituting the calibrated $\tilde{K}$ and $\tilde{\mathbf{p}}_i^s$ into (4), and then, compensating the output $\mathbf{B}_{ai}^s$ by $\tilde{\mathbf{O}}_i$, we can generate data that are closer to the real-world measurement.

III. METHODOLOGY FOR ROBUST MAGNETIC TRACKING

A. MagRobustNet-Based Framework for Robust Tracking

Fig. 3(a) depicts the MagRobustNet-based framework with anomaly detection and measurement recovery for robust magnetic tracking. The proposed framework can detect and indicate the interfered regions, offering an effective means for proactive movement away from interference sources. In addition, our framework possesses the ability to restore anomalous measurements to their expected values when interference is unavoidable.

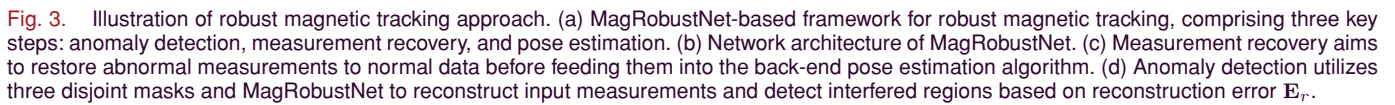
The specific workflow of the proposed framework is as follows. First, the measurements obtained from the sensor array undergo anomaly detection, yielding the reconstruction error $\mathbf{E}_r$. Then, we compare the $\max(|\mathbf{E}_r|)$ with an adaptive threshold defined in (16) to determine whether the input measurements contain anomalous components or not. If it includes anomalies, a corresponding binary mask is generated for MagRobustNet to recover expected measurements of interfered regions from neighboring normal measurements. On the other hand, if it does not contain any anomalous components, the raw measurements are directly fed to the pose estimation algorithms, such as the LM-based tracking method and learning-based PoseNet.

B. Network Architecture of MagRobustNet

UNet is a popular autoencoder network that has achieved remarkable performance in image segmentation. The classical UNet consists of four critical operations: convolution, downsampling, upsampling, and concatenation. However, the conventional UNet and its variants [26] are primarily designed for image-related tasks and are not directly applicable to MTSs, in which the measurement resolution is much lower than that of images.

Therefore, we elaborately designed the UNet-like network named MagRobustNet for our study, as shown in Fig. 3(b), which possesses the following characteristics.

- 1) Partial convolution operators replace standard convolution operators to reduce the impact of invalid data on the output of the convolution operation.
- 2) Downsampling is implemented by setting the padding number of the convolution operator to zero rather than through pooling layers or setting the stride size.



In the convolution operation, the invalid data primarily originates from the padding operation and masking operation. Notably, the proportion of invalid data is surprisingly high for the ultralow resolution measurements of MTSs. For example, given a 5×5 input measurement with 1-pixel padding on each side, $(7 \times 7 - 5 \times 5)/(7 \times 7) = 48.98\%$ of the data will be from padding. These invalid data can cause sharp spikes or dips in filter response values, leading to spatial bias in the feature map and affecting the model performance [27]. Partial convolution was initially proposed by Liu et al. [28] to reduce the impact of invalid input data on the output of convolution operation. Here, $\mathbf{W}$ and b denote the convolution filter weights and bias, respectively, while $\mathbf{I}_{[w,h]}$ and $\mathbf{M}_{[w,h]}$ represent the input data and corresponding binary mask for the current convolutional sliding window. The operation of partial convolution at each position can be defined as

The common downsampling operations, such as the pooling layer and strided convolution, are not employed in MagRobustNet due to the MTS’s low and odd measurement resolution from a 5×5 magnetometer array. For example, with a convolution stride of 2, the resolution of the MagRobustNet feature map in the downsampling path will be 5×5 , 3×3 , 2×2 , and 1×1 , respectively. However, finding a corresponding inverse operation to output odd resolutions during the upsampling path

$$\mathcal{L}_r = \|Net(\mathbf{I} \odot \mathbf{M}_w) - \mathbf{I}\|_1, (w < N) \quad (10)$$

where $\mathbf{M}_w$ is a binary mask that randomly masks out w from N magnetometers. MagRobustNet trained by (10) then, possesses the ability of measurement recovery.

Measurement recovery: As depicted in Fig. 3(c), if the input measurements $\mathbf{I}$ contain anomalies, a corresponding binary mask $\mathbf{M}_w$ is constructed by setting anomalous regions to 0 and valid regions to 1. Supposing $\mathbf{I}_o$ is output of the MagRobustNet, the final reconstructed measurements $\mathbf{I}_r$ are

$$\mathbf{I}_r = \mathbf{I} \odot \mathbf{M}_w + \mathbf{I}_o \odot \bar{\mathbf{M}}_w \quad (11)$$

where $\bar{\mathbf{M}}_w$ is the binary inverse of $\mathbf{M}_w$. Consequently, $\mathbf{I}_r$ contains both the raw normal measurements $\mathbf{I} \odot \mathbf{M}_w$ and the recovered measurements $\mathbf{I}_o \odot \bar{\mathbf{M}}_w$.

However, a key challenge is how to determine the interfered regions. The pixel-level reconstruction methods based on autoencoders and GANs are widely adopted in anomaly detection due to their intuitive nature and outstanding performance [23], [24]. Considering the requirement of measurement recovery, we adopt the UNet-like MagRobustNet to reconstruct the input measurements for anomaly detection.

Anomaly detection: The reconstruction-based anomaly detection relies on the assumption that the trained MagRobustNet can reconstruct measurements with the same distribution well but not those with interferences. The possibility of reconstructing anomalies is far less likely if the anomaly measurements were not visible to the MagRobustNet. To ensure the reconstructed measurements are derived from neighboring regions rather than taking shortcuts, we employ a masking strategy involving k disjoint masks.

Fig. 3(d) illustrates the anomaly detection process based on k disjoint masks, which is described as follows.

- 1) Randomly generate k disjoint binary masks $\mathbf{M}(i)$ such that each measurement cell is masked exactly once. These masks will satisfy the following mathematical expression:

$$\frac{1}{k-1} \sum_{i=1}^k \mathbf{M}(i) = \sum_{i=1}^k \bar{\mathbf{M}}(i) = \mathbf{1} \quad (12)$$

where $\bar{\mathbf{M}}(i)$ represents the binary inverse of $\mathbf{M}(i)$.

- 2) Masked measurements $\mathbf{I} \odot \mathbf{M}(i)$, ($i = 1, 2, \dots, k$) are fed into MagRobustNet sequentially, which endeavors to reconstruct the entire measurements by utilizing the unmasked measurements, i.e.,

$$\mathbf{I}_{oi} = \text{Net}(\mathbf{I} \odot \mathbf{M}(i)), (i = 1, 2, \dots, k). \quad (13)$$

- 3) The output measurements $\mathbf{I}_{oi}$ from the network are masked and synthesized into the final reconstructed measurements $\mathbf{I}_{ra}$, i.e.,

$$\mathbf{I}_{ra} = \sum_{i=1}^k (\mathbf{I}_{oi} \odot \bar{\mathbf{M}}(i)). \quad (14)$$

As $\mathbf{I}_{ra}$ is extrapolated from neighboring measurements, excluding the input measurements that are reconstructed, the possibility of reconstructing anomalies is reduced.

- 4) Finally, the reconstruction errors $\mathbf{E}_r$ (difference between the input and output) can be calculated by

$$\mathbf{E}_r = \mathbf{I}_{ra} - \mathbf{I} = \sum_{i=1}^k (\mathbf{I}_{oi} \odot \bar{\mathbf{M}}(i)) - \mathbf{I}. \quad (15)$$

The reconstruction errors are also known as the anomaly scores, with a higher reconstruction error indicating a higher likelihood that the measurement is an anomaly.

MagRobustNet can be trained by minimizing the loss function $\|\mathbf{E}_r\|_1$ or $\mathcal{L}_r$ in (10). The main difference between these two loss functions is that during each backpropagation, the error computed by $\mathcal{L}_r$ is masking only w/N of the input measurements, while the error computed by $\|\mathbf{E}_r\|_1$ is masking all the input measurements once. In our study, we use the disjoint masking method, i.e., $\|\mathbf{E}_r\|_1$, to train MagRobustNet, which forces the MagRobustNet to reconstruct the masked regions using information from the neighboring regions.

IV. EXPERIMENTS AND VERIFICATION

A. Experimental Platform and Datasets Collection

- 1) *Hardware Platform:* We employed a 5×5 magnetometer array to perceive the magnetic induction intensities around the annular magnet (grad: N35, outer radius $r_1 = 5$ mm, inner radius $r_2 = 3.5$ mm, and length $L = 20$ mm). The sensor array consisted of 25 magnetometers (STMicroelectronics LIS3MDL) that were evenly soldered on a printed circuit board (PCB) with a spacing of 40 mm between each magnetometer. The measurement range was set to ± 800 μT in this study.
- 2) *Datasets:* Our datasets consist of a generated dataset (2.9376 million samples, of which 98% were used for training and 2% for testing) and a real-world test dataset, as shown in Fig. 4. It is worth noting that no interference was added during the generation of the dataset, meaning that all samples in the training dataset are normal unlabeled data.
- 3) *Training Details:* MagRobustNet was implemented in Pytorch and trained on the generated training dataset, employing the Adam optimizer with a base learning rate of 1×10^{-4} . The learning rate was reduced using the cosine annealing schedule over 20 training epochs. Furthermore, PoseNet, an end-to-end pose estimation network proposed by [12], was also trained on the same generated dataset.

B. Assessment of Model-to-Real Domain Adaptation

This experiment primarily aimed to assess the efficacy of the proposed model-to-real domain adaptation approach using the 271 normal samples from the real-world test dataset.

Fig. 5(a) depicts the mean absolute errors (MAEs) between the measurements generated by the uncalibrated and calibrated mathematical models and the actual measurements captured from the sensor array, with values of 1.72 and 0.99 μT , respectively. The t-SNE of samples from the three data domains is presented in Fig. 5(b). Training PoseNet on the dataset generated by the uncalibrated mathematical model achieves an average position accuracy of 2.35 ± 1.48 mm within the height range of 43–153 mm, as shown in Fig. 5(c). After undergoing the pre-calibration operation, the localization accuracy of PoseNet improves to 1.85 ± 1.04 mm. The localization performance of PoseNet closely aligns with the widely adopted optimization-based LM method. These results demonstrate that the

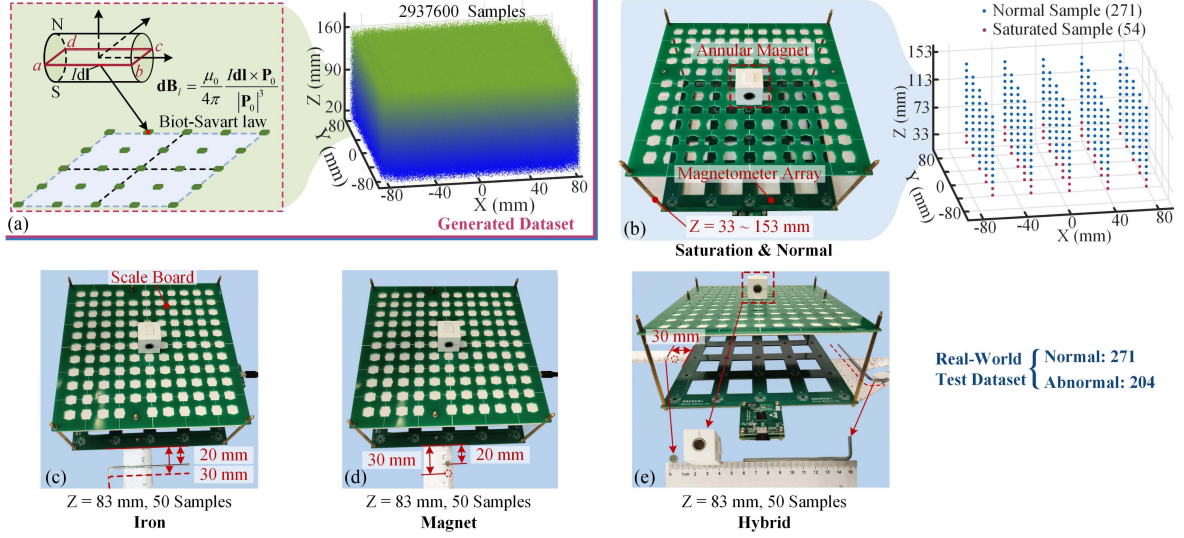


Fig. 4. Overview of our datasets. (a) Generated training dataset was sampled from the prior mathematical model across the working space. (b) Samples were collected at various heights, including 271 normal samples and 54 saturated samples (abnormal samples). (c) Collecting abnormal samples under the interference of a ferromagnetic object (M2 hexagon key with 90 mm length). (d) Taking abnormal samples under the interference of a small magnet ($\Phi \times H = 6 \text{ mm} \times 1.9 \text{ mm}$). (e) Gathering abnormal samples in the presence of hybrid interference arising from a ferromagnetic object and a small magnet. Samples from (b)–(e) together constitute a real-world test dataset with 271 normal samples and 204 abnormal samples.

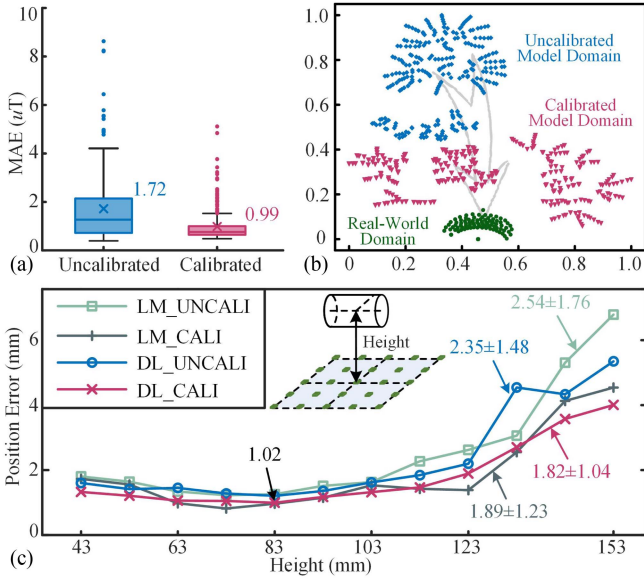


Fig. 5. Evaluating the effectiveness of model-to-real domain adaptation. (a) MAE between the data generated by the mathematical model before and after calibration, and the 271 normal samples from the real-world test dataset. (b) t-SNE of samples from the three domains, illustrating that the data generated by the calibrated model is closer to the real-world domain. (c) Comparison of the position accuracy before and after calibration for pose estimation algorithms based on LM and deep learning (DL, PoseNet) methods.

precalibration operation can promote the transfer of generated data from the model domain to the real-world domain.

C. Anomaly Detection

The magnetic flux density produced by permanent magnets is inversely proportional to the third power of the distance. Our datasets were acquired within the height range of 20–160 mm, so

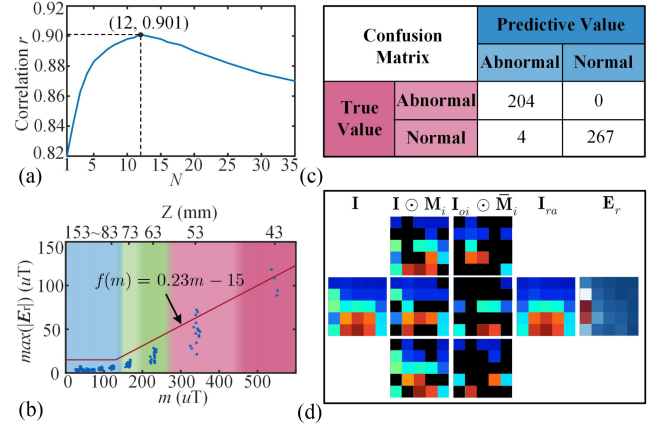


Fig. 6. Adaptive threshold determination and anomaly measurements detection. (a) Pearson correlation coefficient r for different N , where N denotes the number of maxima taken from each sample. (b) Correlation plot between the average of the maximum N values and the maximum reconstruction error, where the red line indicates the adaptive threshold function. Note that the samples in the figure are all normal, and the samples at the height of 33 mm are not included due to sensor saturation. (c) Confusion matrix tested on the real-world test dataset. (d) Snapshots showing the process of anomaly measurement detection.

the measurement values varied widely, i.e., from approximately 0 uT to $\pm 18\,000$ uT. This results in significant variations in reconstruction errors for measurements at different heights. Thus, it is difficult to distinguish between normal and abnormal samples using a fixed threshold.

We observed a positive correlation between the maximum reconstruction error $\max(|E_r|)$ and the average of the maximum N values of input measurements $\bar{I}$. Fig. 6(a) exhibits the Pearson correlation coefficient r for various N . To get a higher correlation, we took $N = 12$.

Further, we used a piecewise linear function to fit the relationship between $\max(|E_r|)$ and the average of the maximum

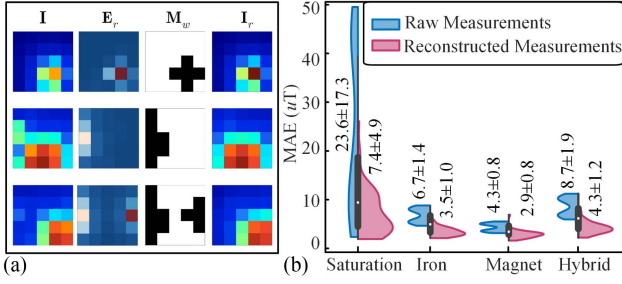


Fig. 7. Performance evaluation of measurements recovery. (a) Several snapshots show the process of generating a mask using the adaptive threshold and recovering the desired measurements. (b) Comparison of MAE metrics before and after performing measurement recovery on abnormal samples.

N values. The bias term of the piecewise linear function was adjusted according to the acceptable false positive rate (FPR) on the normal data. In this study, we set the acceptable FPR to 0.015, which corresponds to the allowance of misclassifying four normal samples as abnormal. The adjusted function, which we refer to as the adaptive threshold function, is shown as the red line in Fig. 6(b) and can be formulated as

$$m = \sum_{i=1}^N \text{top}(I, N) / N \quad (16)$$

$$T(m) = \max(0.23m - 15, 20)$$

where m is the average of the maximum N values from each input I . It is worth highlighting that the adaptive threshold function was determined solely based on normal samples.

Fig. 6(c) presents a confusion matrix tested on the real-world test dataset using the adaptive threshold of (16). MagRobustNet achieved excellent results with a recall rate of 1.0, a precision of 0.981, and an F1 score of 0.990. These metrics indicate that our model effectively captures anomalies, maintains a low false positive rate, and demonstrates well-balanced performance. Alternatively, if we assume an acceptable FPR of zero, MagRobustNet would classify all normal samples correctly but misclassify one abnormal sample as normal, resulting in a recall rate of 0.995, a precision of 1.0, and an F1 score of 0.998.

In both threshold cases, the misclassified samples predominantly occur at heights of 43 and 53 mm, where both saturated and unsaturated samples coexist at the critical saturation boundary. The performance of anomaly detection is influenced by both the chosen thresholds and the dataset itself.

In our proposed framework, we prioritize a threshold with high recall, i.e., the threshold corresponding to an FPR of 0.015. This choice is motivated by the consideration that misclassifying a normal sample as anomalous merely results in an additional measurement recovery step, whereas, failing to detect an abnormal sample will directly affect the subsequent pose estimation algorithm.

D. Measurement Recovery

Fig. 7(a) illustrates generating a mask using the adaptive threshold and recovering the expected measurements. Further, we utilized the MAE metric to quantitatively assess the recovery performance of MagRobustNet on abnormal samples, where the ground truth was generated from the prior mathematical model.

The results in Fig. 7(b) show that the reconstructed measurements are closer to the ground truth, as indicated by the smaller MAE. When the interference source is iron, magnet, or hybrid (iron + magnet), the MAE distribution of the raw measurements shows two peaks corresponding to the interference sources placed at 20 and 30 mm from the magnetometer array during data collection. In contrast, the reconstructed measurements exhibited only one peak, also reflecting the effectiveness of MagRobustNet.

E. Localization Accuracy Evaluation

Our proposed framework is not restricted to learning-based pose estimation networks but can also be seamlessly applied to conventional optimization-based tracking methods. To validate the superiority of the MagRobustNet-based framework, we conducted a comparative analysis of five tracking methods in four interference scenarios, as depicted in Fig. 8. The five tracking methods considered are as follows.

1) LM and 2) PoseNet, where LM refers to the widely adopted LM-based tracking approach [5], [6], [7], and PoseNet is the learning-based pose estimation network proposed by [12]. Both LM and PoseNet are basic tracking methods that do not incorporate anti-interference strategies.

3) MagRobustNet+LM and 4) MagRobustNet+PoseNet, which utilize our MagRobustNet-based framework with LM and PoseNet as the pose estimation algorithms.

5) HuberLoss+LM [16], an anti-interference method based on the Huber loss and LM algorithm. To the best of the authors' knowledge, HuberLoss+LM is the only previously proposed solution to cope with the interferences targeted in this study.

The experimental results in Fig. 8 illustrate that the localization accuracy of the LM and PoseNet methods is significantly improved in four interference scenarios after applying the MagRobustNet-based framework. Specifically, the MagRobustNet-based framework reduces the average position error of PoseNet by 76.2% (10.18 to 2.42 mm) and the average orientation error by 68.8% (from 12.69° to 3.96°) across the four scenarios. The improved localization performance (2.42 mm and 3.96°) has significant practical implications for various applications, including WCE tracking [1], AGV navigation [3], and body motion capture [4].

Moreover, the MagRobustNet+PoseNet method exhibits superior anti-interference performance compared to the MagRobustNet+LM method, possibly attributed to the fact that solving the magnetic tracking problem with the LM algorithm is based on the assumption of Gaussian distribution. In contrast, PoseNet and MagRobustNet were trained on the same dataset, following the same data distribution and underlying patterns. Thus, the measurements recovered by MagRobustNet better align with the expected input of PoseNet.

It should be noted that our proposed framework shows no advantage over the HuberLoss+LM approach in scenarios with iron interference. The performance difference is due to their distinct mechanisms for handling interference.

Our proposed framework is based on measurement recovery. When the number of interfered sensors increases, the area requiring recovery becomes larger, leading to a decrease in localization accuracy. The HuberLoss+LM approach, on the other hand, reduces the weight of outlier measurements using the Huber loss function, which behaves linearly for outliers and quadratically for normal measurements [16]. Consequently,

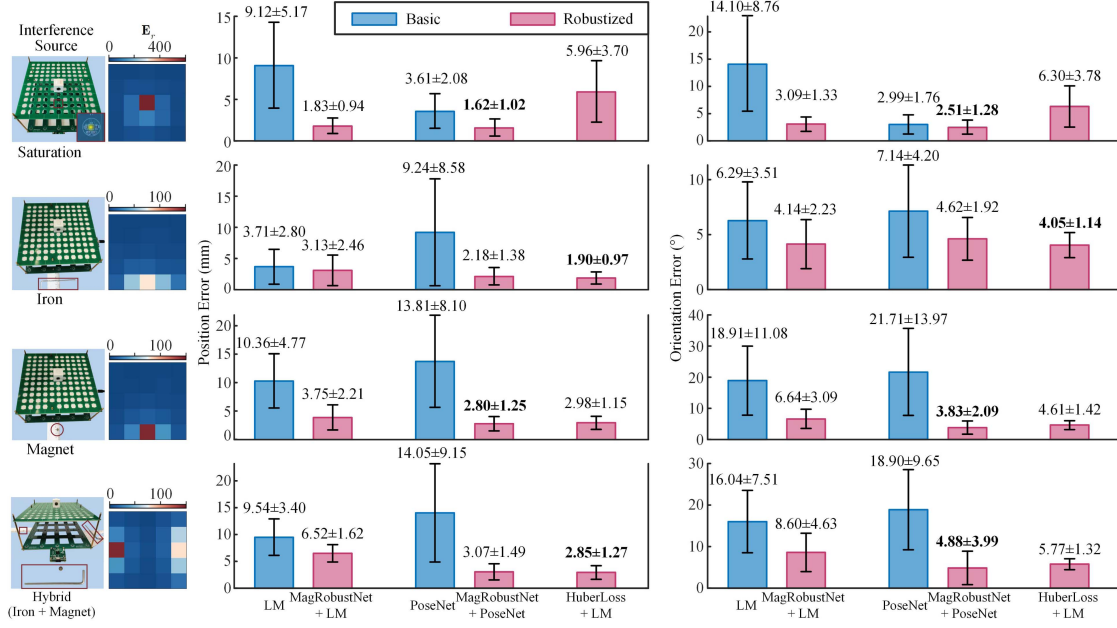


Fig. 8. Localization accuracy comparison of different methods under four interference scenarios. The localization accuracy of the methods based on our proposed framework (MagRobustNet+LM and MagRobustNet+PoseNet) is significantly improved over the conventional LM and PoseNet methods. When compared to the HuberLoss+LM approach, MagRobustNet+PoseNet demonstrates comparable performance in mitigating the effects of soft and hard magnetic interferences (iron, magnet, hybrid) while exhibiting superior capabilities in handling sensor saturation.

TABLE I
MAGROBUSTNET PERFORMANCE WITH DIFFERENT MASKING METHODS AND MASKING RATIOS

	Interference Type	Disjoint Masking Method				Conventional Masking Method		
		$k = 2$	$k = 3$	$k = 4$	$k = 5$	$w = 12$	$w = 8$	$w = 4$
Position Error (mm)	Saturation	1.79 ± 1.10	1.62 ± 1.02	1.48 ± 0.76	1.65 ± 0.92	1.91 ± 0.93	2.13 ± 1.15	2.72 ± 2.80
	Iron	2.56 ± 1.81	2.18 ± 1.38	2.57 ± 1.88	3.44 ± 1.80	2.30 ± 1.44	2.43 ± 1.89	2.97 ± 1.71
	Magnet	2.77 ± 3.39	2.80 ± 1.25	2.94 ± 1.27	3.11 ± 1.20	2.91 ± 1.41	2.83 ± 1.23	3.28 ± 1.38
	Hybrid	3.16 ± 1.58	3.07 ± 1.49	3.21 ± 1.65	5.28 ± 2.42	2.95 ± 1.48	3.12 ± 1.45	5.13 ± 2.05
	Avg.	2.57 ± 1.97	2.42 ± 1.29	2.55 ± 1.39	3.37 ± 1.59	2.52 ± 1.32	2.63 ± 1.43	3.53 ± 1.99
Orientation Error (°)	Saturation	2.61 ± 1.29	2.51 ± 1.28	2.33 ± 1.10	2.47 ± 1.15	2.66 ± 1.47	2.57 ± 1.48	2.74 ± 2.14
	Iron	5.23 ± 2.36	4.62 ± 1.92	4.29 ± 1.79	5.14 ± 2.63	5.19 ± 2.04	5.08 ± 1.85	4.67 ± 2.52
	Magnet	3.69 ± 2.21	3.83 ± 2.09	3.85 ± 2.22	4.26 ± 2.03	4.11 ± 2.61	3.65 ± 2.34	4.34 ± 2.75
	Hybrid	5.15 ± 4.09	4.88 ± 3.99	4.72 ± 3.65	4.93 ± 3.41	5.27 ± 3.90	4.64 ± 3.43	5.01 ± 3.41
	Avg.	4.17 ± 2.49	3.96 ± 2.32	3.85 ± 2.19	4.20 ± 2.31	4.31 ± 2.51	3.99 ± 2.28	4.19 ± 2.71

when the interference source is iron, which introduces smaller interference magnitudes over a larger area, the HuberLoss+LM method outperforms our framework. Conversely, when the interference source is saturation, which introduces larger interference magnitudes over a smaller area, our proposed framework significantly outperforms the HuberLoss+LM method. Despite not always outperforming the HuberLoss+LM method, our framework offers following advantages.

- 1) *Versatility*: It has broader applicability, being suitable for both deep learning-based and optimization-based tracking methods, whereas the HuberLoss+LM method is only applicable to optimization-based methods.
- 2) *Enhanced Functionality*: In addition to improving localization accuracy similar to the HuberLoss+LM method, it can also indicate interfered regions, providing a means to actively avoid interference.

F. Ablation Study

In this study, we trained MagRobustNet using disjoint mask sets to mask the input measurements. However,

another conventional practice is randomly masking w out of the 25 magnetometers, as shown in (10). Table I shows the experimental results of the trained model based on disjoint mask sets and the traditional masking method. To ensure a fair and reasonable comparison, each model applied an identical mask to the same abnormal sample during inference to reconstruct its expected measurements.

The results from Table I indicate that different values of k demonstrate their respective advantages in the four interference scenarios. For instance, when the interference type is a magnet, MagRobustNet performs optimally with $k = 2$. On balance, we selected $k = 3$ to ensure favorable performance across four types of interferences. In this case, the mask set consists of three disjoint binary masks, two of which have a masking ratio of 8/25, and the other one is 9/25.

In addition, the results in Table I validate that the disjoint masking method outperforms the conventional masking method. The superior performance of the disjoint mask sets method is mainly attributed to its strategy to calculate the loss after all input measurements are masked, rather than only masking part of the input measurements during each backpropagation, as shown in

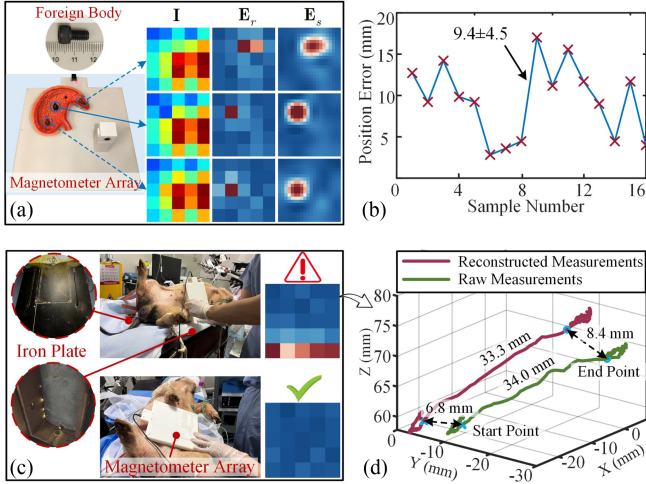


Fig. 9. Demonstration of clinical applications in a phantom and in vivo experiments. (a) Snapshots of the experimental scene and measurements, where E_s is derived from the reconstruction error E_r after bicubic interpolation. (b) Positioning errors of the foreign body at different locations. (c) Indicating the presence of interference sources and prompting the adjustment of the array to an interference-free position. (d) Motion trajectories of the WCE before and after performing measurements recovery.

(10). In contrast, the disjoint masking method uses (13)–(15) to calculate the loss, which enforces MagRobustNet to reconstruct expected measurements from neighboring data.

G. Demonstration of Clinical Applications

Foreign body ingestion in children is a common occurrence in clinical practice. As shown in Fig. 9(a), we conducted a foreign body positioning experiment based on the reconstruction error E_r (also known as score map) on a stomach model. The experiment involved both the annular magnet and a foreign body, making it challenging to observe where the foreign body was directly from the raw measurements I . However, the score map allows us to visualize the location of the foreign body intuitively. Since the original score map resolution was only 5×5 , bicubic interpolation was applied to increase the resolution to 20×20 . We uniformly collected 16 samples above the magnetometer array and used the interpolated score map for position estimation. The positioning accuracy is shown in Fig. 9(b). The results demonstrate that our method offers a new potential diagnostic means for detecting and locating ingested foreign bodies in clinical practice, though currently limited to ferromagnetic materials.

In addition, detecting and indicating interfered regions is crucial for achieving high-quality and effective outcomes in practical applications. In vivo experiment shown in Fig. 9(c), our proposed framework successfully detected the presence of an interference source (a metal plate), prompting the adjustment of the array to an interference-free location. The metal plate was located in the middle of the operating table, which is often overlooked as a potential interference source. Even in cases where the MTS cannot avoid interference, our method can significantly enhance the localization performance by recovering interfered measurements. Fig. 9(d) compares the motion trajectories before and after performing measurements when the WCE moves about 30 mm under interference. The results indicate that the

interference source, an iron plate in this case, mainly produces bias errors for localization. Moreover, the errors are nonlinear and vary with the position of the WCE (the distance between the two start points is 6.8 mm, and the two endpoints are 8.4 mm apart). Thus, the proposed method can help ensure accurate and reliable tracking results.

V. CONCLUSION

In this study, we propose a MagRobustNet-based framework with anomaly detection and measurement recovery to enhance the anti-interference of MTSs. Experiments on the real-world test dataset showed a 76.2% improvement in average position accuracy across four interference scenarios. Compared to the existing HuberLoss + LM method, the proposed framework exhibits broader applicability to both deep learning-based and optimization-based methods, and enhanced functionality in indicating interfered regions. In addition, our method offers a new potential diagnostic means for detecting and locating ingested foreign bodies in clinical practice.

A limitation of our proposed method is its inability to address interferences that can disturb all magnetometers, such as geomagnetic fields, as our approach relies on recovering the expected measurements of interfered regions from neighboring normal data. In future research, we will further address the situation where all magnetometers are subject to interference to improve the anti-interference capability and application scope of MTSs.

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